

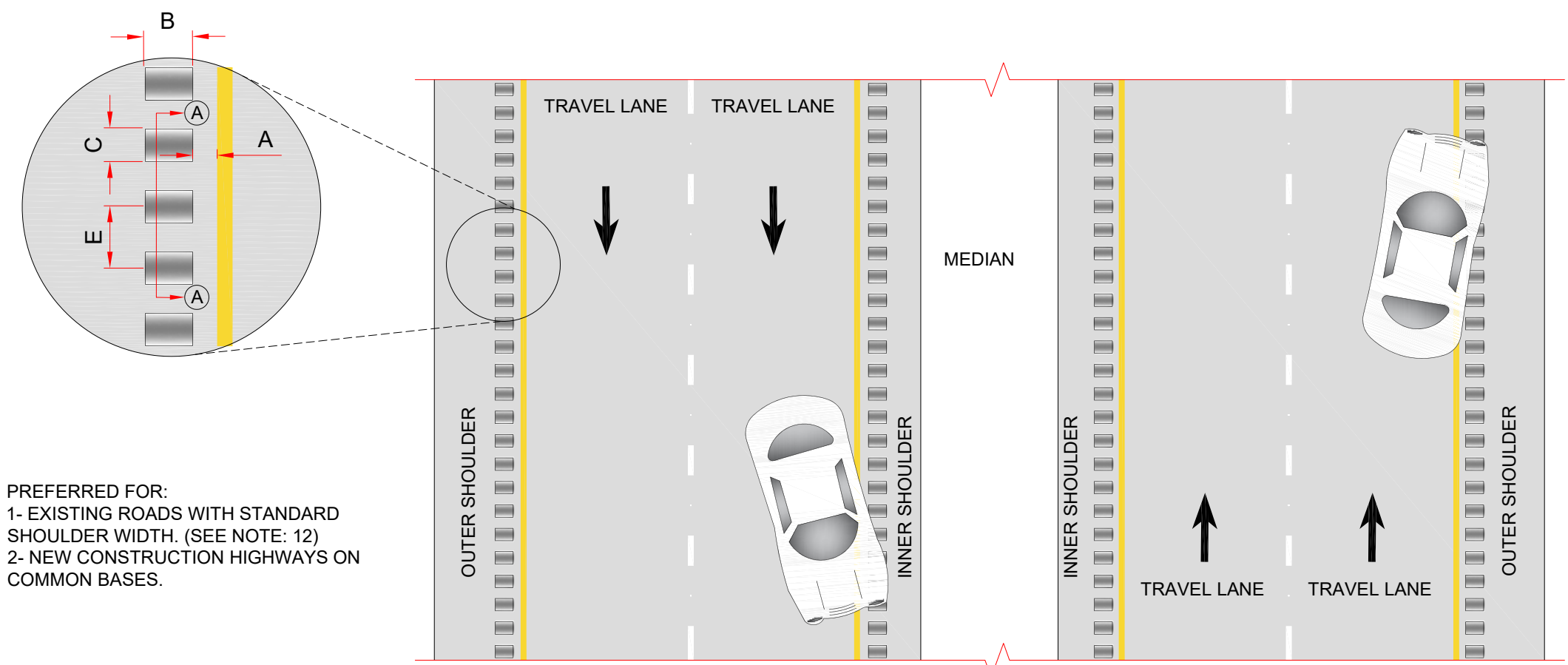


FIGURE (1): SHOULDER RUMBLE STRIPS PATTERN ON DIVIDED & SINGLE HIGHWAYS (CONTINUOUS APPLICATION) N.T.S

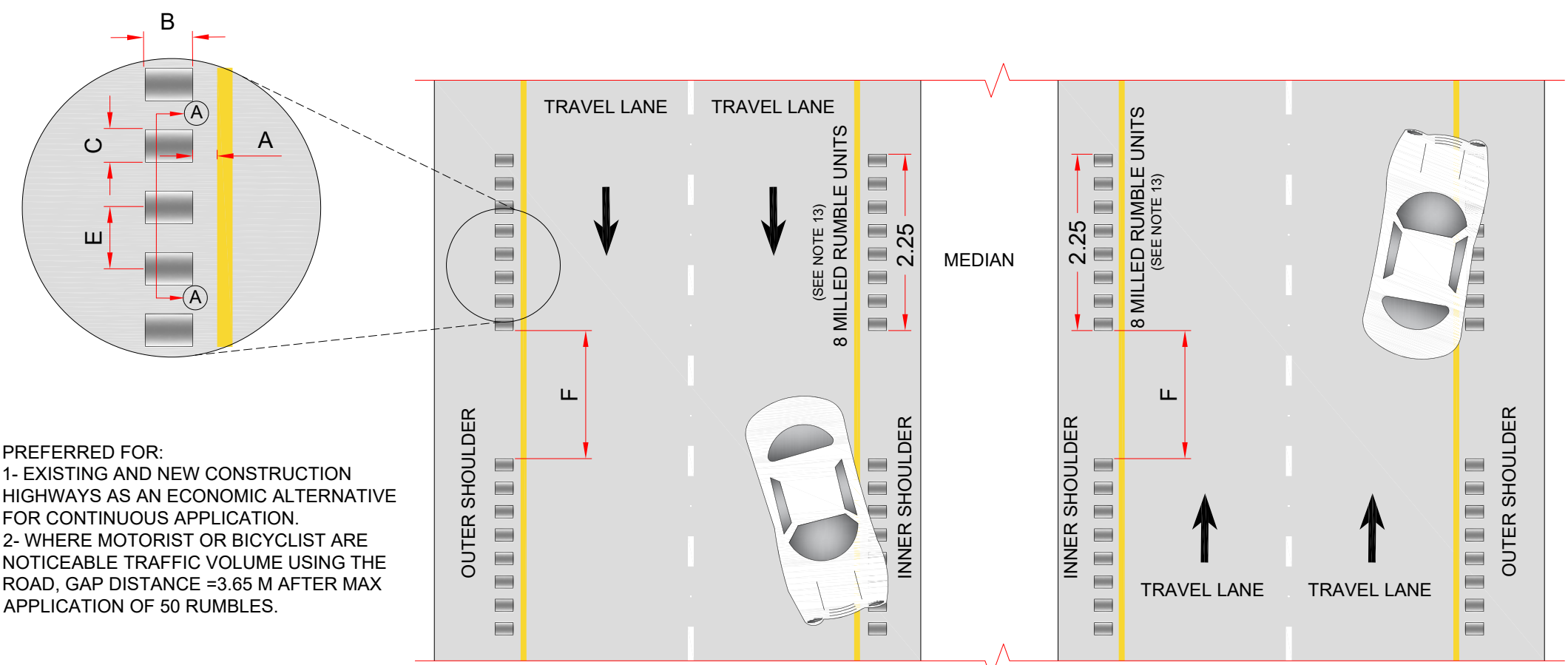


FIGURE (3): TYPICAL APPLICATION PATTERN OF RUMBLES WITH GAP DISTANCE ON DIVIDED & SINGLE HIGHWAY SHOULDERS (APPLICABLE FOR RUMBLE STRIPS & RUMBLE STRIPES TYPES) N.T.S

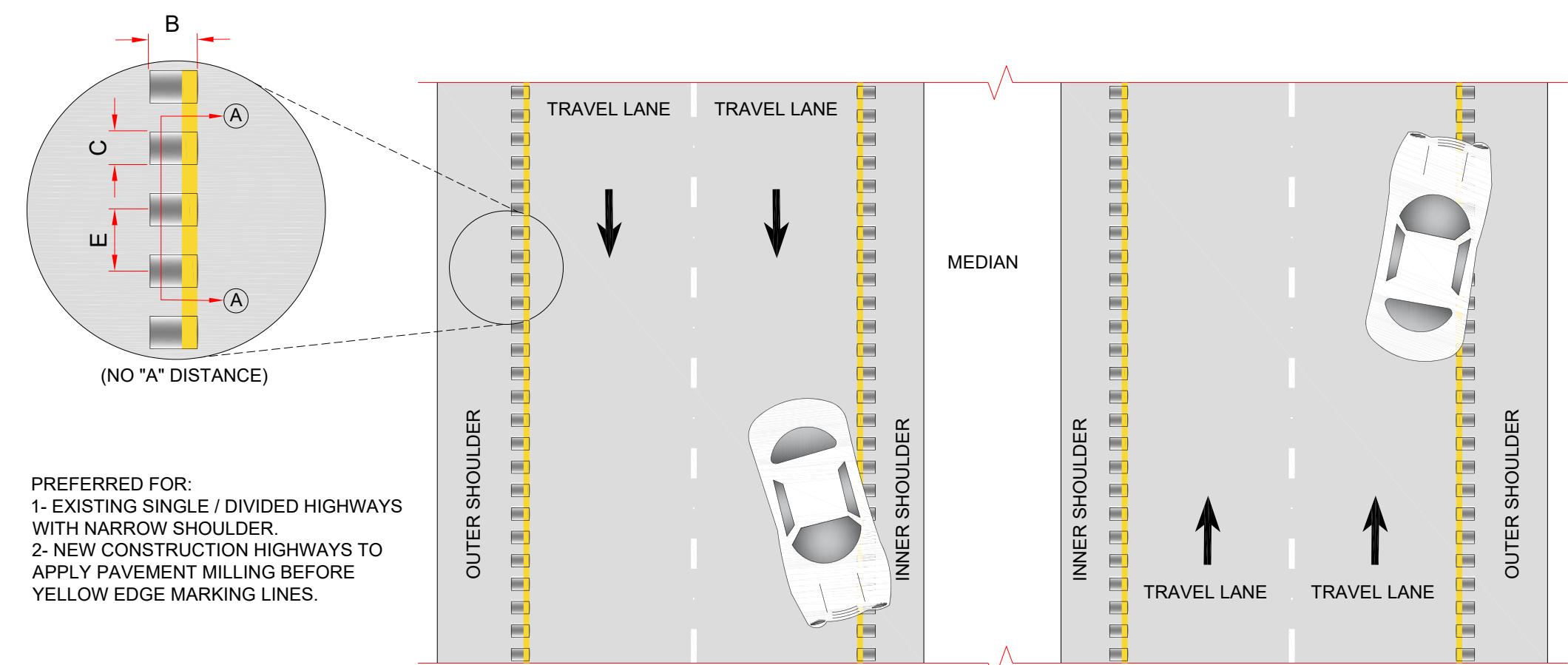


FIGURE (5): SHOULDER RUMBLE STRIPES PATTERN ON DIVIDED & SINGLE HIGHWAYS (CONTINUOUS APPLICATION) N.T.S

OFFSET	LENGTH	WIDTH	DEPTH	SPACING	GAP DISTANCE
A	B	C	D	E	F
300 MM	350 MM	150 MM	12 MM	300 MM	1.50 M (REPEATED AFTER EVERY 8 RUMBLE UNITS)

TABLE (1): TYPICAL RUMBLE STRIP / STRIPE UNIT DIMENSIONS AND SPACING

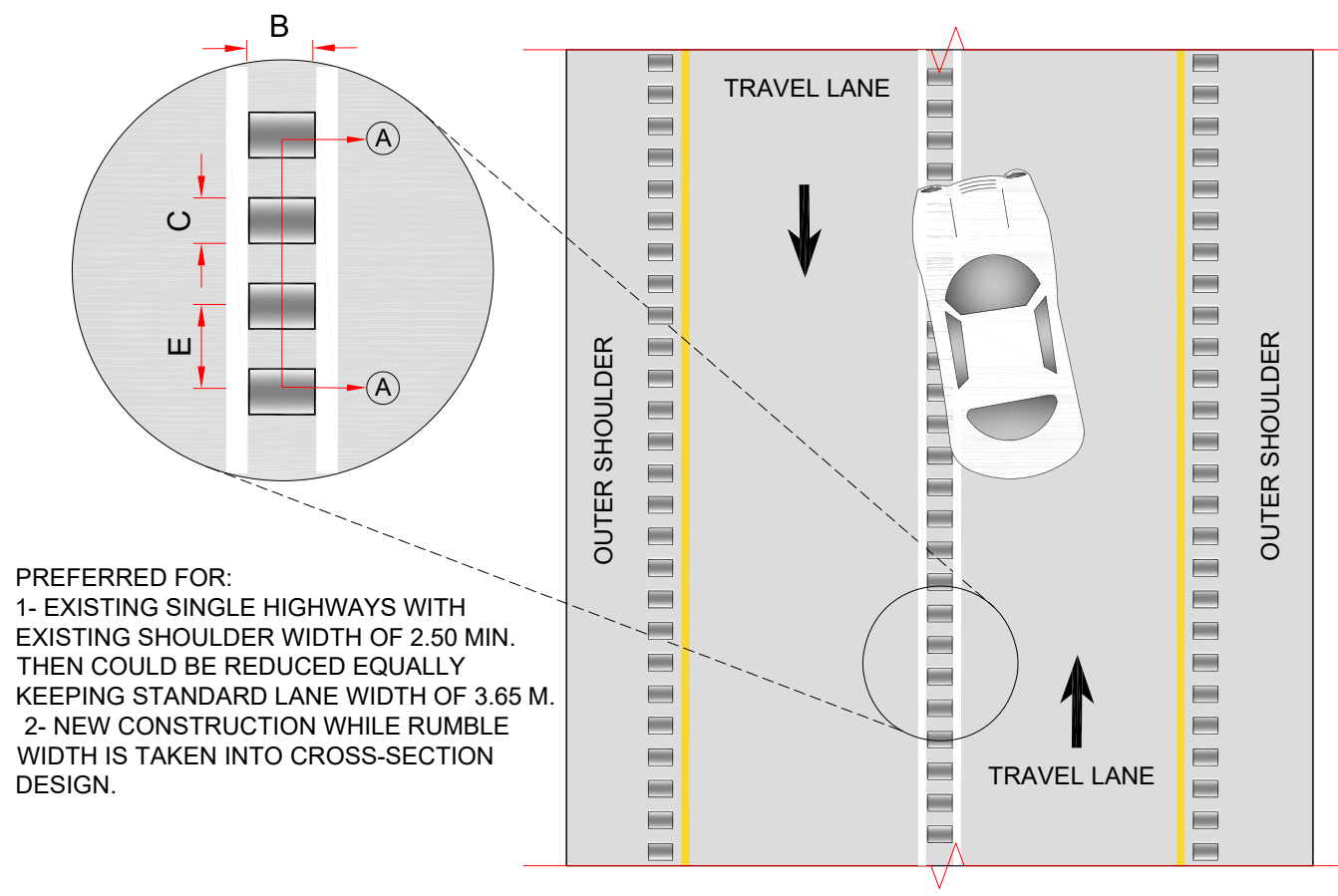


FIGURE (2): CENTER LINE RUMBLE STRIPS PATTERN ON SINGLE HIGHWAY (CONTINUOUS APPLICATION) N.T.S

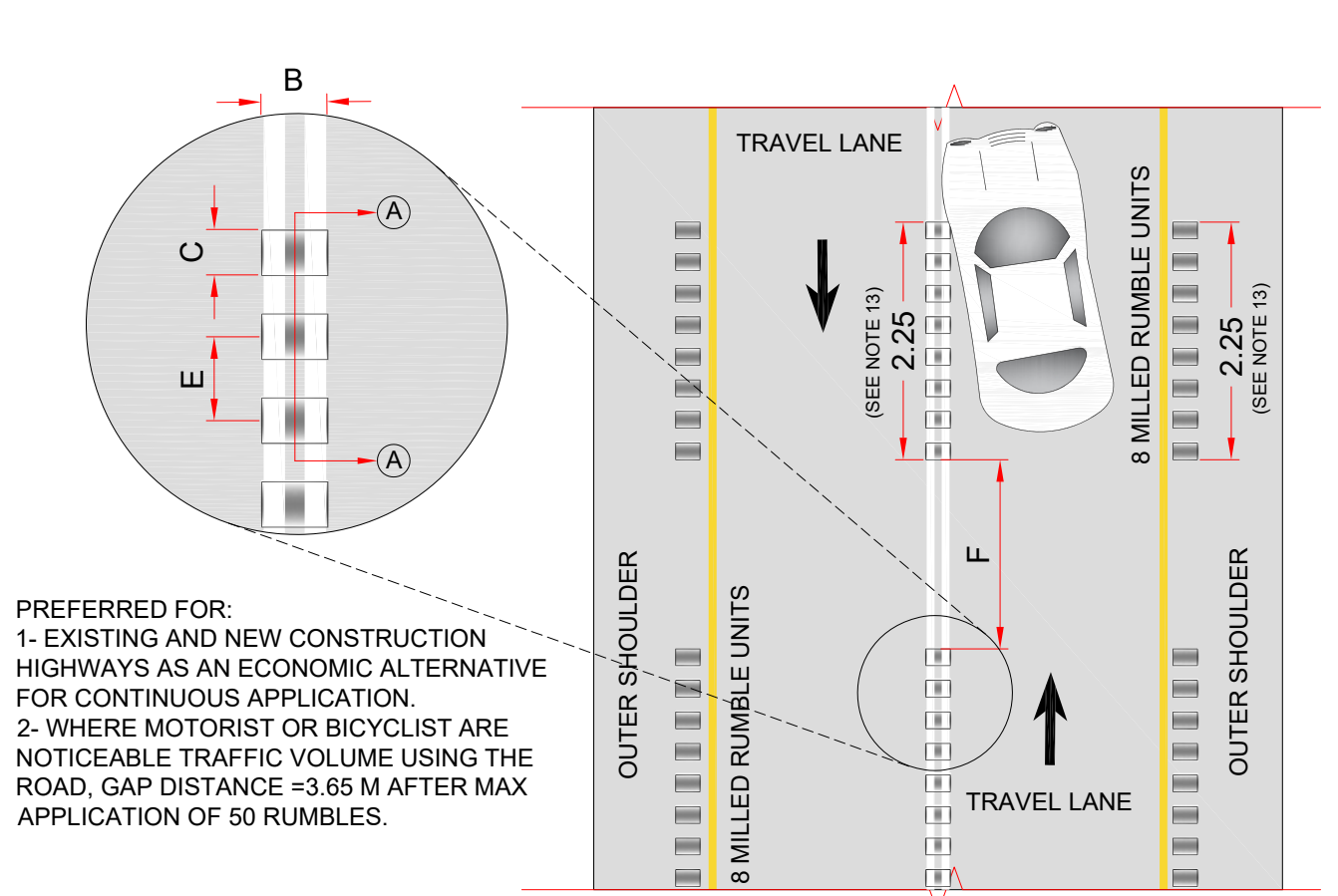


FIGURE (4): TYPICAL APPLICATION PATTERN OF RUMBLES WITH GAP DISTANCE ON SINGLE HIGHWAY CENTERLINE (APPLICABLE FOR RUMBLE STRIPS & RUMBLE STRIPES TYPES) N.T.S

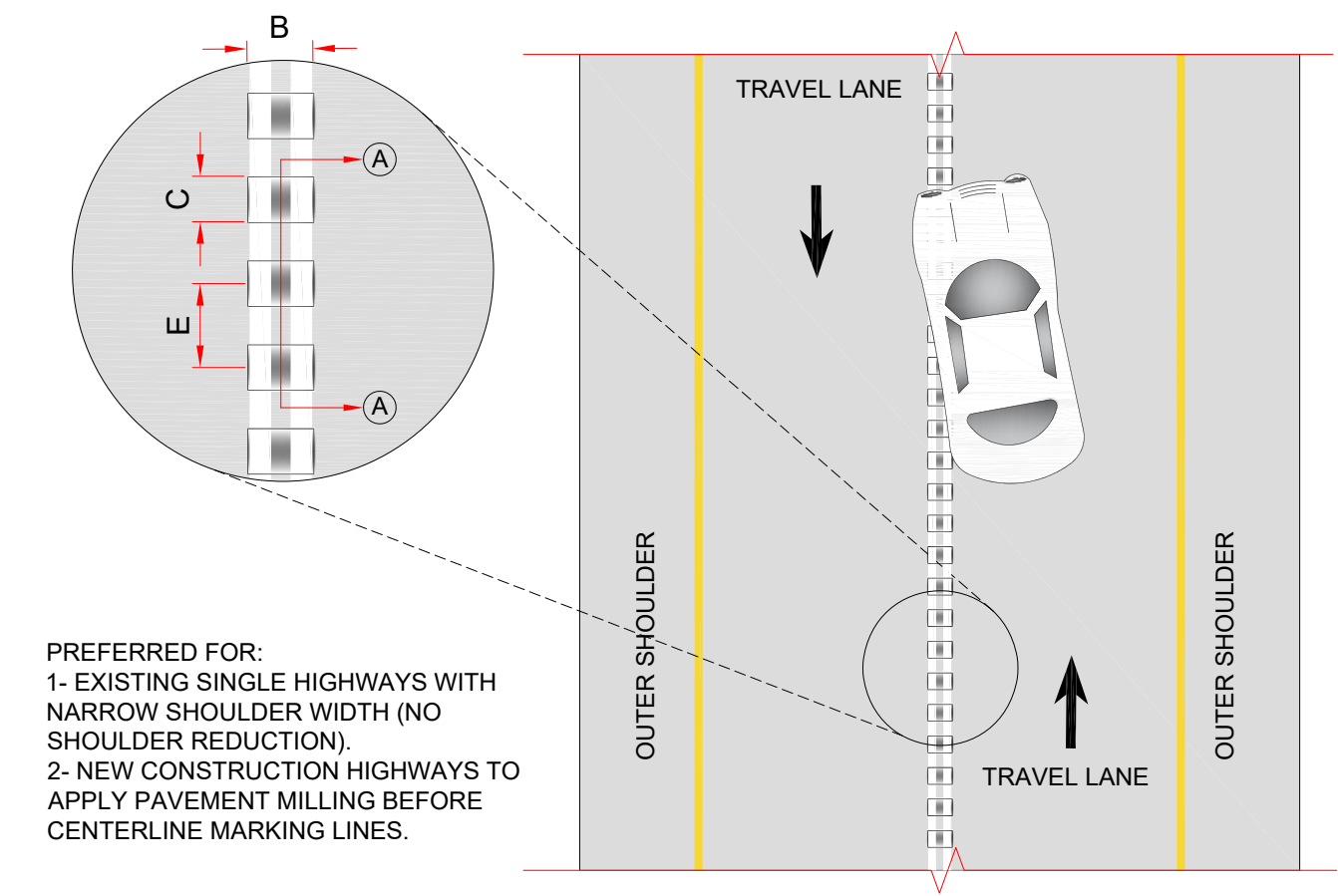
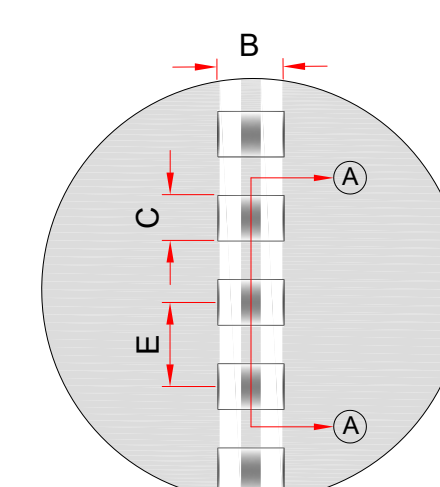
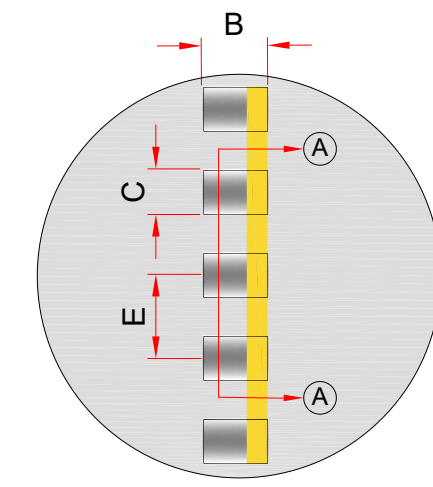
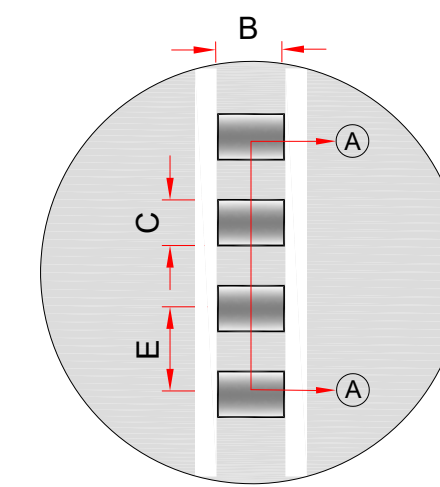
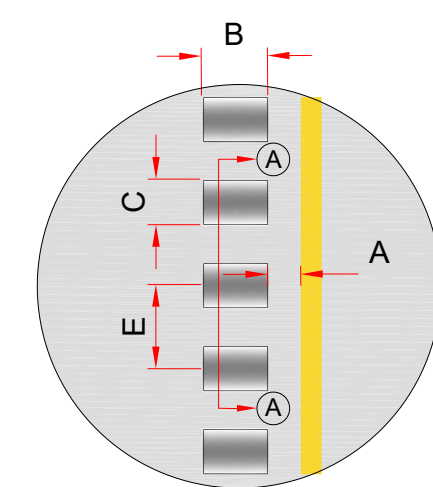
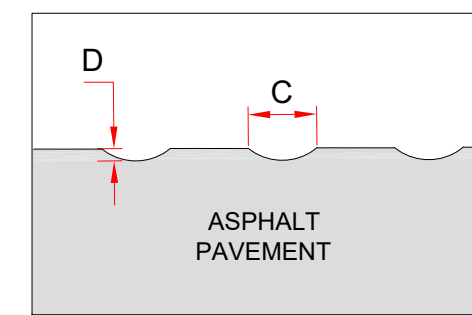
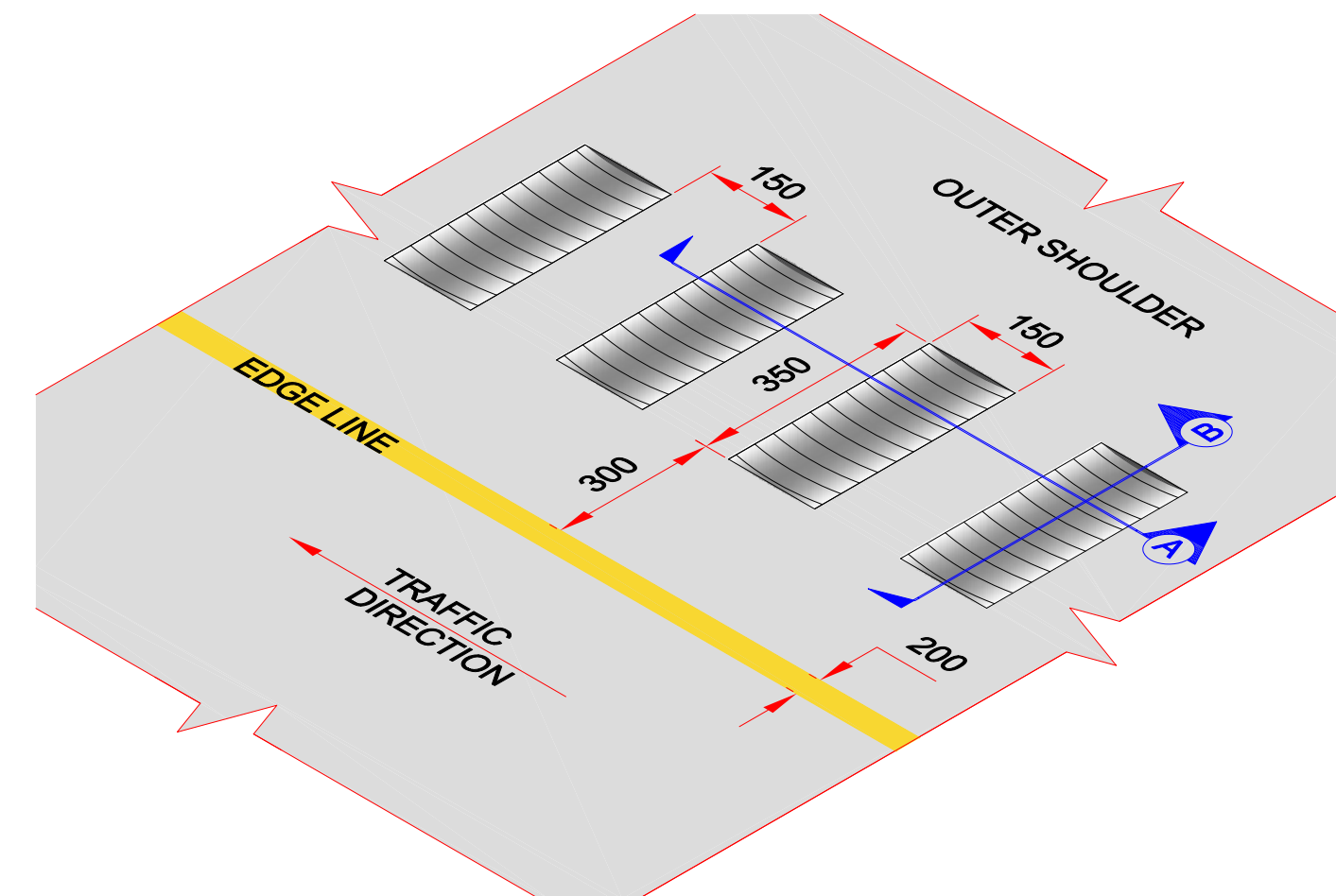


FIGURE (6): CENTER LINE RUMBLE STRIPES PATTERN ON SINGLE HIGHWAY (CONTINUOUS APPLICATION) N.T.S

SHOULDER WIDTH VALUE REQUIRED TO INSTALL RUMBLE STRIPE (METER)		
SHOULDER & HIGHWAY TYPE	DIVIDED HIGHWAY	SINGLE HIGHWAY
OUTER SHOULDER	DESIRABLE 3 M	DESIRABLE 2.5 M
	MINIMUM 2.5 M	MINIMUM 2 M
INNER SHOULDER	2 M	N.A

TABLE (2): MINIMUM PAVED SHOULDER WIDTH REQUIRED TO INSTALL RUMBLE STRIPS / STRIPES



NOTES:

- ALL DIMENSION SHOWN ARE IN MILLIMETER UNLESS OTHER WISE MENTIONED
- A RUMBLE STRIP BECOMES A "RUMBLE STRIPE" WHEN AN EDGE LINE OR CENTER LINE PAVEMENT MARKING IS PLACED ON THE RUMBLE STRIP.
- INDICATE ON THE CONSTRUCTION PLANS THE RECOMMENDED RUMBLE STRIP LOCATIONS AND GAPS.
- FOR RUMBLE STRIPES, MILL THE RUMBLE STRIPS PRIOR TO THE INSTALLATION OF PAVEMENT MARKINGS.
- PROVIDE A GAP IN THE CENTER LINE RUMBLE STRIP INSTALLATION AT PUBLIC ROAD INTERSECTIONS AND MAJOR APPROACH ROADS.
- CENTER LINE RUMBLE SIZES COULD BE SIMPLY DECREASED TO FIT WITH MARKING LINES WIDTH REQUIREMENTS.
- PROVIDE A GAP IN THE SHOULDER RUMBLE STRIP INSTALLATION AT THE FOLLOWING LOCATIONS:**
 - (3.65 M) FOR BICYCLE / MOTORISTS USERS AFTER A MAXIMUM (14.85 M = 50 RUMBLES) CONTINUOUS INSTALLATION.
 - AT LEAST (5 M) PRIOR TO THE BEGINNING OF THE TAPER FOR A RIGHT TURN DECELERATION LANE OR SERVICE ROAD EXITS.
 - AT LEAST (23 M) PRIOR TO A PULLOUT, MAILBOX TURNOUT, OR RADIUS RETURN OF AN INTERSECTION WITHOUT A DECELERATION LANE.
 - AT LEAST (5 M) BEYOND A PULLOUT, MAILBOX TURNOUT, OR RADIUS RETURN OF AN INTERSECTION.
 - AT LEAST (7.5 M) PRIOR TO A MINOR APPROACH, SUCH AS A DRIVEWAY.
 - AT LEAST (5 M) BEYOND A MINOR APPROACH.
- THE PLACEMENT OF SHOULDER RUMBLE STRIPS CAN BE SITE-SPECIFIC OR PROJECT-WIDE. EXAMPLES OF APPROPRIATE INSTALLATION LOCATIONS INCLUDE:**
 - AT HIGH INCIDENT ROADWAY DEPARTURE (RWD) CRASH SITES.
 - AT LIMITED CLEAR ZONE AREAS SUCH AS: STEEP SLOPES, TREES, UTILITIES, ROCK OUTCROPS, SHARP BENDS, ETC.
 - ON HORIZONTAL CURVES.
 - ON LONG TANGENT SECTIONS.
 - AT APPROACHES TO A NARROW BRIDGE.
- THE PLACEMENT OF CENTER LINE RUMBLE STRIPS CAN BE SITE-SPECIFIC OR PROJECT-WIDE. EXAMPLES OF APPROPRIATE INSTALLATION LOCATIONS INCLUDE:**
 - AT HIGH INCIDENT HEAD-ON AND OPPOSING-SIDESWIPE CRASH SITES.
 - ALONG ROADWAYS WITH FREQUENT HORIZONTAL CURVATURE PREFERABLY THROUGH A CONTINUOUS AREA; OR LESS COMMONLY AT SPECIFIC SPOT LOCATIONS WITH ISSUES.
 - IN PASSING AND NO-PASSING ZONES.
- AVOID THE INSTALLATION OF RUMBLE STRIPS AND STRIPES AT THE FOLLOWING LOCATION :**
 - ON BRIDGE DECK AND APPROACH SLABS.
 - RIGHT TURN ACCELERATION AND DECELERATION LANES
 - PAVEMENT WEARING SURFACE LESS THAN (50MM) IN DEPTH.
 - AFTER PAVEMENT MARKINGS HAVE BEEN PLACED.
 - WHEN PAVEMENT CONTAINS ANY CRACKING TYPE, OR GENERALLY IN POOR CONDITION HENCE, CONSULT WITH THE PAVEMENT ENGINEER TO VERIFY A SUITABLE PAVEMENT CONDITION PRIOR TO MILLING PROCESS.
- REFER TO MOT HDM -VOL.2 -SECTION 1.06 -TABLE 1.06.1 -SHOULDER WIDTH.
- CONTRACTORS SHOULD FOLLOW THE SHOWN DIMENSIONS IN THIS DRAWING UNLESS HAVING A WRITTEN APPROVAL FROM MOT SAFETY ENGINEER TO USE OF DIFFERENT RUMBLE DIMENSIONS BEFORE APPLICATION.
- APPLICATION OF 8 OR 9 MILLED RUMBLES AS SHOWN IN FIGURES 3 & 4 IS SUBJECT TO MOT ENGINEER APPROVAL KEEPING THE SAME GAP DISTANCE OF 1.50 M.

KINGDOM OF SAUDI ARABIA
MINISTRY OF TRANSPORT



TITLE :

المخطط النموذجي لتفاصيل الإمتزازات التحذيرية المكشوفة
في سطح الرصف الاسفلتي

TYPICAL DETAILS OF MILLED RUMBLE STRIPS / STRIPES ON HIGHWAYS PAVEMENT SURFACE

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